DATA ANALYTICS USING PYTHON – PRACTICAL EXAMINATION

Duration: 2 Hours

Maximum Marks: 100

Instructions:

- Write efficient and readable Python code.
- Use functions wherever appropriate.
- Use object-oriented programming wherever required.
- Avoid using built-in shortcuts unless explicitly permitted.
- Write comments wherever necessary.

SECTION A – Python Programming (60 Marks)

Question 1 (10 Marks)

Create a Python program to manage employee information using classes.

Your program should allow users to store employee details, calculate the annual salary, and identify employees whose annual salary exceeds ₹8,00,000.

Question 2 (10 Marks)

Given a list of student records stored as tuples containing student names and marks, write a Python program to:

- Find the highest-scoring student.
- Calculate the average marks.
- Display all students scoring above the average.
- Store the topper's details in a dictionary.

Question 3 (10 Marks)

A retail store maintains product-wise monthly sales data using a dictionary where each key represents a product and the corresponding value is a list of monthly sales.

Write a program to:

- Calculate the total sales of each product.
- Determine the highest-selling product.
- Compute the average sales of every product.
- Store the final summary in another dictionary.

Question 4 (10 Marks)

Given a paragraph of text, write a Python program without using the `collections` module to:

- Count the total number of words.
- Identify all unique words.
- Calculate the frequency of every word.
- Display the three most frequently occurring words.

Question 5 (10 Marks)

Write a menu-driven Python application for a list of integers that provides the following operations:

- Remove duplicate elements.
- Display all prime numbers.
- Display all even numbers.
- Display all odd numbers.
- Find the largest and smallest elements.
- Exit the application.

Implement each operation using separate functions.

Question 6 (10 Marks)

Design a **Bank Account Management System** using classes.

The program should support:

- Depositing money.
- Withdrawing money.
- Checking the available balance.
- Maintaining a transaction history.
- Preventing invalid withdrawal and deposit operations.

SECTION B – Exploratory Data Analysis (40 Marks)

Use the dataset provided by the examiner.

Question 7 (15 Marks)

Load the dataset using Pandas and perform the following exploratory tasks:

- Display the first five records.
- Display the last five records.
- Display the shape of the dataset.
- Display the data types of all columns.
- Identify missing values.
- Identify duplicate records.
- Remove duplicate records if present.

Question 8 (15 Marks)

Perform descriptive statistical analysis on the dataset.

Your program should compute:

- Mean
- Median
- Mode
- Standard Deviation
- Correlation Matrix

Based on your analysis, answer the following:

- Which feature has the strongest positive correlation with the target variable?
- Which feature has the weakest correlation with the target variable?
- Which numerical feature exhibits the highest variance?

Question 9 (10 Marks)

Using appropriate visualization libraries, generate the following plots:

- Histogram
- Box Plot
- Scatter Plot
- Correlation Heatmap

Based on the generated visualizations, answer the following:

- Identify any outliers present in the dataset.
- Comment on the distribution of the numerical features.
- Identify any pair of highly correlated features.
- State which visualization is most suitable for detecting outliers and justify your answer.

BONUS QUESTION (10 Marks)

An e-commerce company stores customer order details as a list of dictionaries.

Write a Python program to generate a customer-wise summary containing:

- Total number of orders placed.
- Total purchase amount.
- Average purchase amount.
- Customer with the highest expenditure.

Your solution should make effective use of dictionaries, loops, functions, and classes wherever appropriate.
